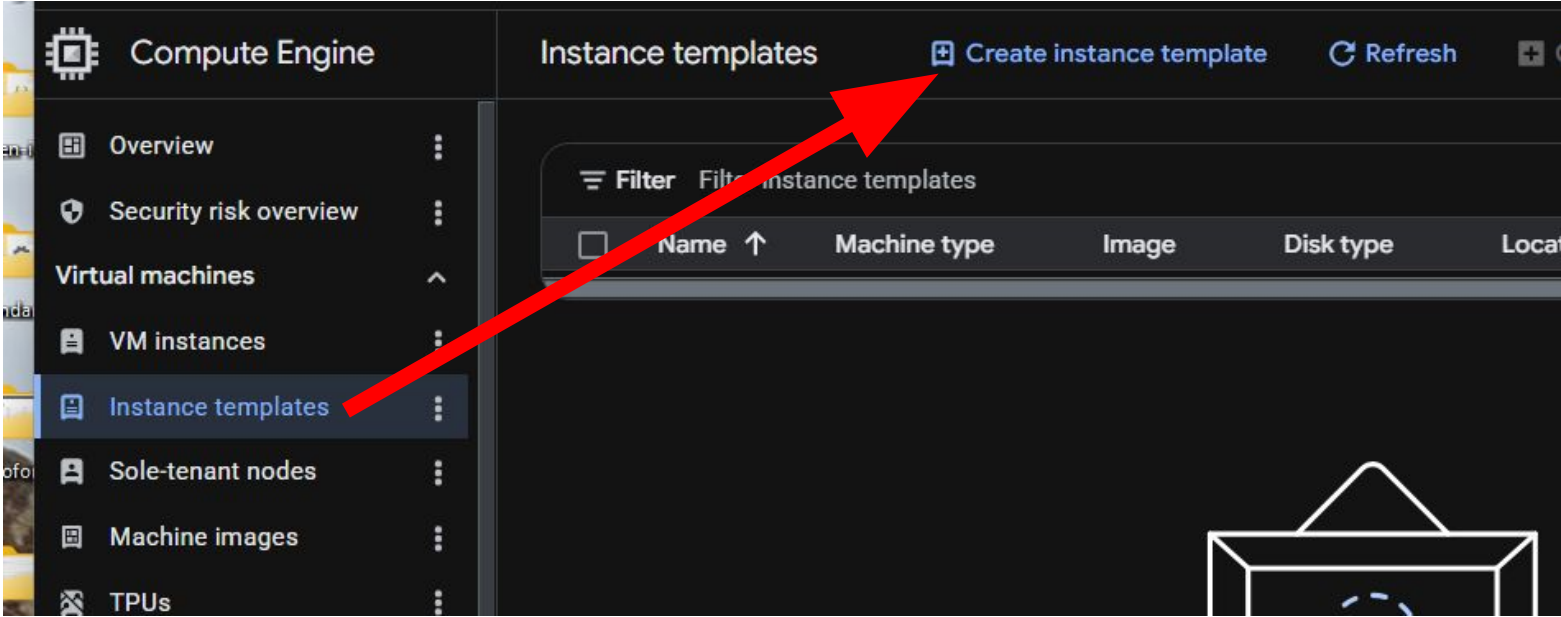


Runbook

Goal: to create fully configure external application global load balancer in clickops, with a MIG as the backend. And showing how to connect the load balancer up to a MIG.

Click on create instance template



Compute Engine

Overview

Security risk overview

Virtual machines

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Committed use discou...

Reservations

Migrate to Virtual Mac...

← Create an instance template

Describe a VM instance once and then use that template to create groups of identical instances [Learn more](#)

Name *

globals-lab

▼ Manage tags and labels

Location

Global instance templates can be used in any region. To lower the impact of outages outside your region and gain data residency within your region, use a regional instance template. [Learn more](#)

☐ Regional (recommended)

☒ Global

Generate machine type suggestions with Gemini

Make sure Global is selected (if you don't you are stuck on the one region)

- Click on Advanced options and go to Management -> Automation -> Startup Script add the script you are planning to use.

The screenshot shows the 'Management' tab in the Google Cloud console, which includes sections for Reservations, Automation, and Metadata. The 'Automation' section is active, showing a startup script that has been entered into a text area. The script is as follows:

```
Startup script
echo "Try:"
echo " curl -s localhost/healthz"
echo " curl -s localhost/metadata | jq "
```

Below the script, there is a link to 'Learn more' about startup scripts. The 'Metadata' section below it explains how to set custom metadata for an instance. At the bottom, there is a section for 'Customer Managed Encryption Key (CMEK) revocation policy' with two options: 'Shut down (recommended)' and 'Do nothing' (which is selected).

Management
Description, deletion protection, reservations, and automation

Description

Reservations

- ☒ Use automatic selection
Google Cloud will select an existing reservation that matches properties of your instance template
- ☐ Choose a reservation
- ☐ Don't use a reservation

Automation

Startup script

```
echo "Try:"
echo " curl -s localhost/healthz"
echo " curl -s localhost/metadata | jq "
```

You can choose to specify a startup script that will run when your instance boots up or restarts. Startup scripts can be used to install software and updates, and to ensure that services are running within the virtual machine. [Learn more](#)

Metadata

You can set custom metadata for an instance or project outside of the server-defined metadata. This is useful for passing in arbitrary values to your project or instance that can be queried by your code on the instance. [Learn more](#)

[+ Add item](#)

Customer Managed Encryption Key (CMEK) revocation policy
You can configure this VM to shut down when the key associated with any attached disk is revoked. [Learn more](#)

- ☐ Shut down (recommended)
- ☒ Do nothing

Instance templates

Create instance templateRefreshCreate VMCreate instance groupCopyDelete

Instance templates are saved VM configurations used to create identical VMs, either individually or as part of managed instance groups. [Learn more](#)

FilterFilter instance templates

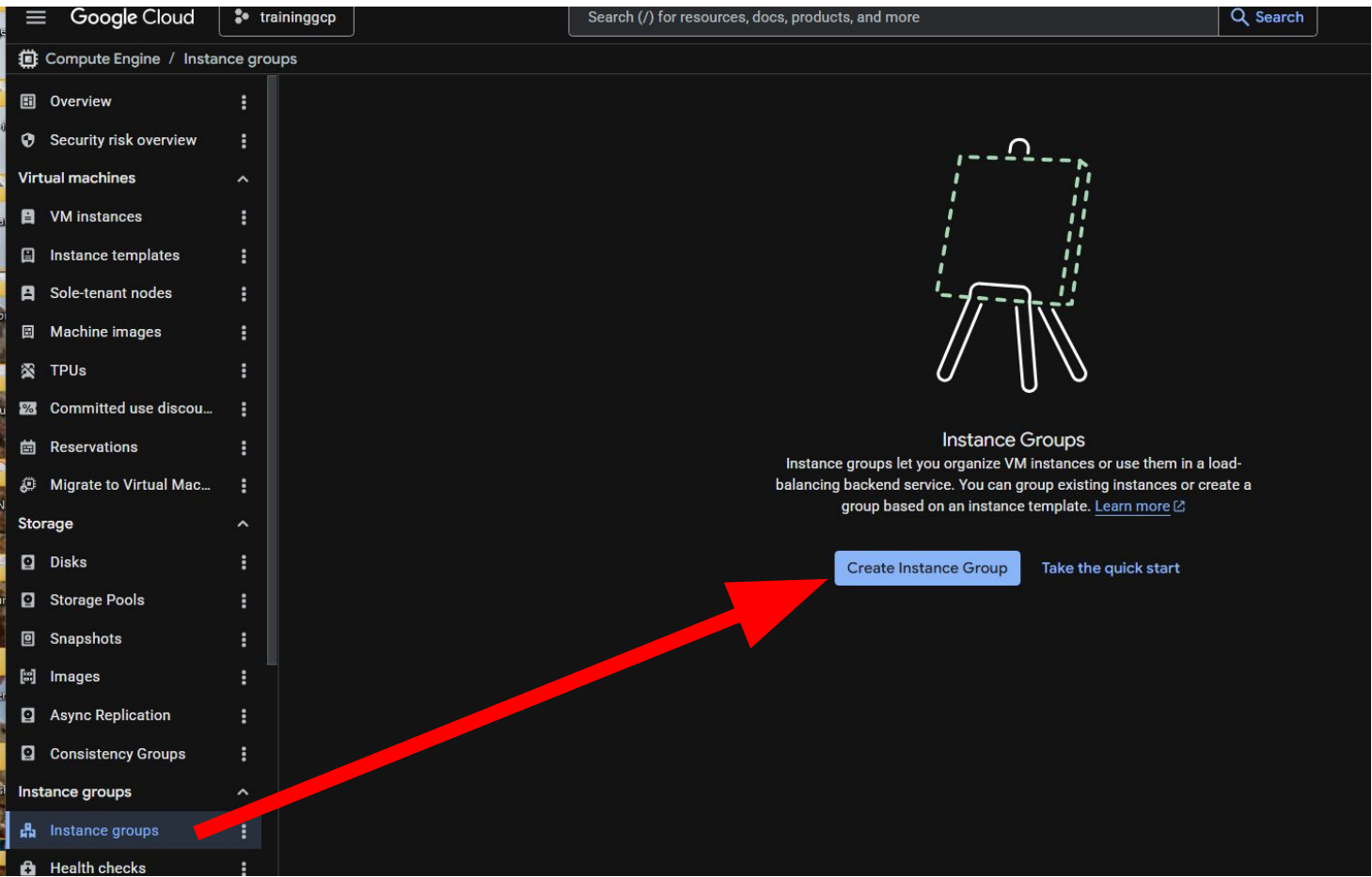
☐	Name ↑	Machine type	Image	Disk type	Location ?	Placement policy ?	Actions
☐	globals-lab	e2-medium	debian-12-bookworm-v20260513	Balanced persistent disk	global	No policy	

SelectPermissions

Instance template done

Section 2 MIG

1. As in the first section go to the 3 hashes in the top left corner go to Compute Engine and Instance groups



Set up automatic management for a group of stateless VMs, including updates, regional deployments, load balancing, autoscaling, and autohealing. [Learn more](#)

Name *

global-mig-lab



Name is permanent

Description

Instance template *



Filter | Type to filter

globals-lab

e2-medium, debian-12-bookworm-v20260513, global

[Create a new instance template](#)

Group after

global-mig-lab



Name is permanent

Description

Instance template *

globals-lab



e2-medium, debian-12-bookworm-v20260513, global



You've selected a global instance template. To lower impact of outages outside your region, use a regional instance template instead. [Learn more](#)

Number of instances *

4

Location

For higher availability, select multiple zones in a region instead of a single zone. [Learn more](#)

- ☐ Single zone
- ☒ Multiple zones

Region *
us-central1 (Iowa) ▼ ?

Target distribution shape
Balanced

Instance redistribution ?

☐ Allow instance redistribution

Zones

☒	us-central1-c
☒	us-central1-a
☒	us-central1-f
☒	us-central1-b

Cancel OK



Autoscaling

Use autoscaling to automatically add and remove instances to the group for periods of high and low load. [Learn more](#)

Made 4 zones

Autoscaling

Use autoscaling to automatically add and remove instances to the group for periods of high and low load. [Learn more](#)

Autoscaling mode

On: add and remove instances to the group

Minimum number of instances *

3



Maximum number of instances *

9



Autoscaling signals



Use signals to help determine when to scale the group. [Learn more](#)

CPU utilization: 60% (default)



Predictive autoscaling is off

[Add a signal](#)

Autoscaling schedules



Initialization period



Specify how long it takes for your app to initialize from boot time until it is ready to serve. [Learn more](#)

Initialization period *

60

seconds



Scale-in controls



VM instance lifecycle

Configure what happens when VMs are created, repaired, and deleted.

Action on failure

Select the action to take when a VM in your instance group fails. [Learn more](#)

Default action on failure

Repair instance



Autohealing



Autohealing recreates VM instances if your application cannot be reached by the health check.

[Learn more](#)

Health check

load-balanced-vms-autohealing-health-check (HTTP)



Making healthcheck

te instance group

60 seconds ?

Scale-in controls

VM instance lifecycle

Configure what happens when VMs are created, repaired, and deleted.

Action on failure

Select the action to take when a VM in your instance group fails. [Learn more](#)

Default action on failure

Repair instance

Autohealing

Autohealing recreates VM instances if your application cannot be reached by the health check. [Learn more](#)

Health check

Filter Type to filter

No health check

demo-mig-healthcheck-1 (HTTP)

port: 80, timeout: 5s, check interval: 5s, unhealthy threshold: 3 attempts, location: global

healthcheck01 (TCP)

port: 80, timeout: 5s, check interval: 5s, unhealthy threshold: 2 attempts, location: global

load-balanced-vms-autohealing-health-check (HTTP)

port: 80, timeout: 5s, check interval: 5s, unhealthy threshold: 10 attempts, location: global

migonehealthcheck1 (TCP)

port: 80, timeout: 5s, check interval: 5s, unhealthy threshold: 2 attempts, location: us-central1

Create a health check

[firewall rules to allow health checking](#)

Updates during VM instance repair

Name *

mighealthcheck-global

Lowercase, no spaces.

Description

mighealthcheck-global

Scope

☒ Global

☐ Regional

Protocol

TCP

Port *

80

PROXY protocol

None

Request

Response

Logs

☐ On

Turning on Health check logs can increase costs in Logging.

☒ Off

Health criteria

Define how health is determined: how often to check, how long to wait for a response, and how many successful or failed attempts are decisive

Check interval *

5 seconds ?

Timeout *

5 seconds ?

Healthy threshold *

2 consecutive successes ?

Unhealthy threshold *

2 consecutive failures ?

Save Cancel

Default action on failure

Repair instance

Autohealing

Autohealing recreates VM instances if your application cannot be reached by the health check.

[Learn more](#)

Health check

mighealthcheck-global (TCP)

port: 80, timeout: 5s, check interval: 5s, unhealthy threshold: 2 attempts, location: global



Because this health check will replace unhealthy instances, you may want to increase the check interval to at least 10 seconds and the unhealthy threshold to at least 3 times so your instances aren't replaced too often. [Learn more](#)

Initial delay *

300

Seconds



On failed health check

Default action



To use autohealing, configure firewall rules. This will allow the health check to connect to VM instances in the group. [How to configure firewall rules to allow health checking](#)

Updates during VM instance repair

Google Cloud

Cloud Training

Cloud Architecture / Cloud Compute / Instance groups

Cloud

Compute Engine / Instance groups

Overview

Security risk overview

Virtual machines

VM instances

Instance templates

Sole-tenant nodes

Machine images

TPUs

Instance groups

Create instance group

Refresh

Delete

Instance groups are collections of VM instances that use load balancing and automated services, like autoscaling and autohealing. [Learn more](#)

Filter

Enter property name or value

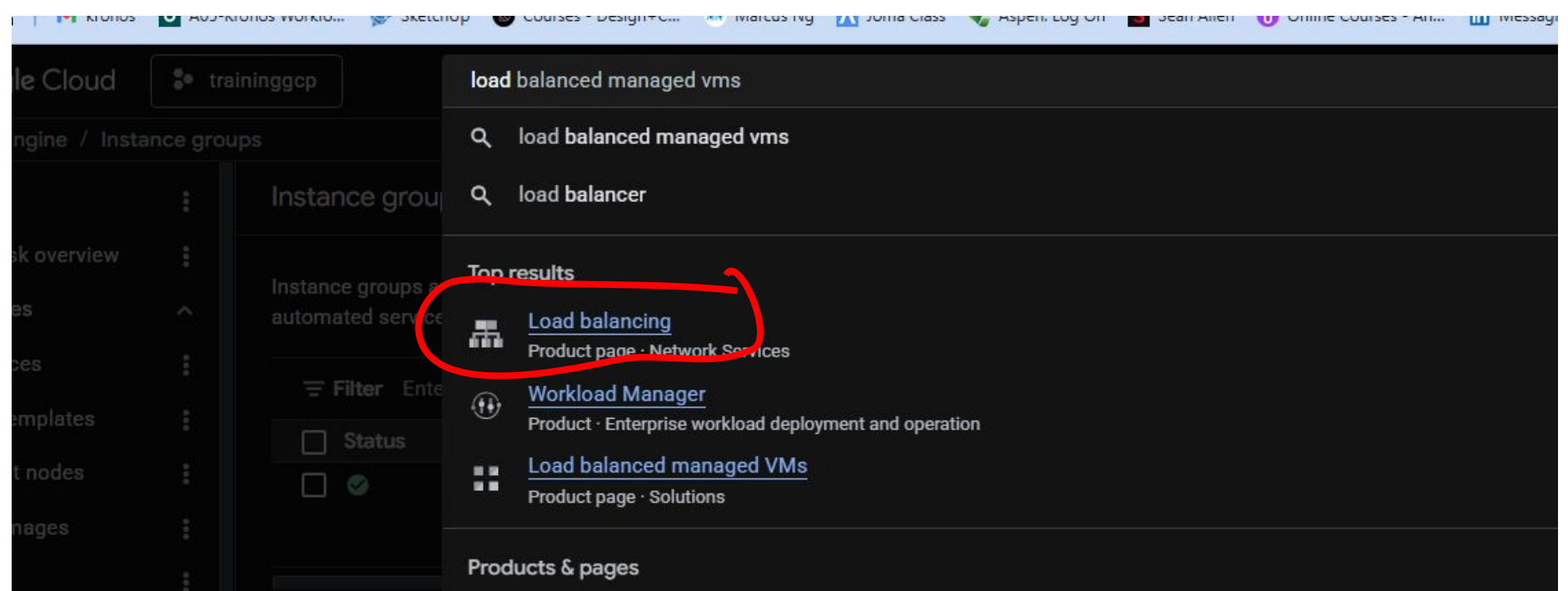
☐	Status	Name ↑	Instances	Template	Group type	Creation time	Recommendation	Autoscaling	Location
☐		global-mig-lab	3	globals-lab	Managed	May 16, 2026, 10:11:39 PM UTC-05:00		No configuration	us-central1 (3/4)

5

6

7

Section 3 Global Load Balancer





Google Cloud



traininggcp

load



Network Services



Load balancing



Cloud DNS



Cloud CDN



Cloud NAT



Cloud Service Mesh (Traf...



Service Directory



Cloud Domains

Load balancing

+ Create load balancer

Refresh

Delete



Monitor + Analyze Load Balanced Traffic

Use VPC Flow Logs and Flow Analyzer to gain flow level insights for your

- ✓ Diagnose connectivity issues
- ✓ Identify top talkers + Analyze network utilization
- ✓ For passthrough load balancers, understand how flows are being dist

[Try now](#)

[Remind me later](#)

Load balancing

Cloud DNS

Cloud CDN

Cloud NAT

Cloud Service Mesh (Traf...

Service Directory

Cloud Domains

Private Service Connect

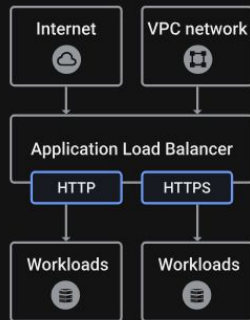
Composite Health

SSL policies

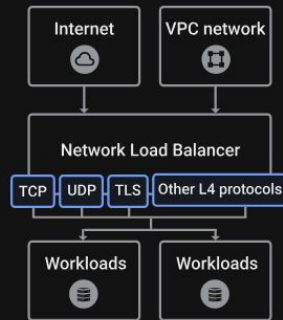
Service Extensions

1 Type of load balancer

- ☒ **Application Load Balancer (HTTP/HTTPS)**
Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic.



- ☐ **Network Load Balancer (TCP/UDP/SSL)**
Choose a Network Load Balancer when you need TLS offloading at scale, support for UDP, and exposing IP addresses to your applications.



Next

2 Public facing or internal

Public facing (external)

3 Global or single region deployment

Global workloads

4 Load balancer generation

Global external Application Load Balancer

5 Create load balancer

Configure

Cancel

Google Cloud

traininggcp

load

Network Services

Load balancing

Cloud DNS

Cloud CDN

Cloud NAT

Cloud Service Mesh (Traf...

Service Directory

Cloud Domains

Private Service Connect

Composite Health

SSL policies

Service Extensions

Create a load balancer

✓ Type of load balancer

Application Load Balancer

✓ Public facing or internal

Public facing (external)

✓ Global or single region deployment

Global workloads

4 Load balancer generation

Global external Application Load Balancer

Load balancer with EXTERNAL_MANAGED load balancing scheme (Recommended)

[Learn more](#)

Classic Application Load Balancer

Previous generation load balancer with EXTERNAL load balancing scheme

[Learn more](#)

External Client

Internet

Global external Application Load Balancer

Advanced Traffic Management

Region A Workloads

Region B Workloads

External Client

Internet

Classic Application Load Balancer

Basic Traffic Management

Region A Workloads

Region B Workloads

Next

5 Create load balancer

Configure

Cancel

Marketplace

Load balancing

Cloud DNS

Cloud CDN

Cloud NAT

Cloud Service Mesh (Traf...

Service Directory

Cloud Domains

Private Service Connect

Composite Health

SSL policies

Service Extensions

✓ Type of load balancer

Application Load Balancer

✓ Public facing or internal

Public facing (external)

✓ Global or single region deployment

Global workloads

✓ Load balancer generation

Global external Application Load Balancer

5 Create load balancer



You are about to create an
Application Load Balancer

with following features:

- Public facing (external)
- Global

Configure

Cancel

Load Balancer name *

Lowercase, no spaces.

✓ Frontend configuration

• Backend configuration

• Routing rules

i Review and finalize (optional)

Backend configuration

Create or select a backend service for incoming traffic. You can also create a new backend service.



Only backend services created for HTTP(S) Load Balancing for the Classic HTTP(S) Load Balancer cannot be used.

Dismiss

Backend services & backend buckets

Filter Type to filter

☐ chica-backend-bucket

Create a backend service

Create a backend bucket

Name *

Lowercase, no spaces.

Description

Backend type

Instance group

Protocol

HTTP

Named port *

http

Timeout *

30

seconds

IP address selection policy

Only IPv4

Health check *

Backends

^ New backend

IP stack type

☒ IPv4 (single-stack)

☐ IPv4 and IPv6 (dual-stack) ?

Instance group *



Monitor + Analyze Load Balanced Traffic

Use VPC Flow Logs and Flow Analyzer to gain flow level insights for your Load Balancers and backends. [Learn more](#)

- ✓ Diagnose connectivity issues
- ✓ Identify top talkers + Analyze network utilization
- ✓ For passthrough load balancers, understand how flows are being distributed among their backends

[Try now](#)

[Remind me later](#)

[Load balancers](#)

[Backends](#)

[Frontends](#)

[Service LB policies](#)



Faster web performance and improved web protection with Cloud CDN and Cloud Armor. [Learn more](#)

[Filter](#) Enter property name or value

☐	Name	Load balancer type	Access type	Protocols	Region	Backends	Actions
☒	load-1bk	Application	External				
☐	nginx-balancer-global	Application	External	HTTP		✓ 1 backend bucket	⋮

To view or delete load balancing resources like forwarding rules and target proxies, go to the [load balancing components view](#).

Lets create cloud armor

Google Cloud

Network Security

Cloud Armor

DDoS Dashboard

Cloud Armor policies

Adaptive Protection

Cloud Armor Service Ti...

Cloud IDS

IDS Dashboard

IDS Endpoints

IDS Threats

Cloud NGFW

Dashboard

Firewall policies

Threats

Firewall endpoints

Secure Access Connect

Realms

Attachments

Cloud Armor policies

Located in this projectInherited by this project

Cloud Armor advanced network DDoS protection is now generally available to protect applications and services using Network Load Balance

Dismiss

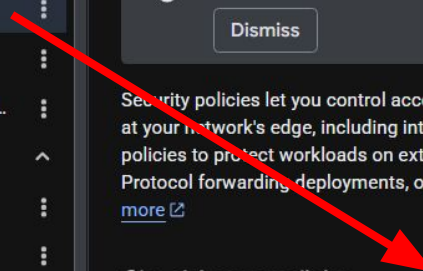
Security policies let you control access to your Google Cloud Platform resources at your network's edge, including internal Load Balancers. You can use security policies to protect workloads on external Cloud Load Balancing deployments, Protocol forwarding deployments, or Instances with public IP addresses. [Learn more](#)

Cloud Armor policies

Create policyRefresh

Filter Enter property name or value

☐	Name ↑	Type	Scope	Rules	Targets
☐	default-security-policy-for-backend-service-chica-backend	Backend security policy	global	2	0
☐	default-security-policy-for-backend-service-global-backend	Backend security policy	global	2	0
☐	default-security-policy-for-backend-service-seco-friday-backend	Backend security policy	regional (us-central1)	2	0



Cloud traininggcp Search (/) for resources, docs, products, and more

Security

Create security policy

Description

cloud-armor-week9

Policy type

☒ Backend security policy

☐ Edge security policy

☐ Network edge security policy

Scope

☒ Global

☐ Regional

Policy layer

☒ Application

☐ Network

Default rule action

Action *
Deny

Response code *
403 (Forbidden)

Next step

- Add more rules (optional)
- Apply policy to targets (optional)

Targets are Google Cloud resources that you want to control access to. You can use external load balancer backend services as targets.

Targets

Type 1

Backend service (external application load b...

Backend service 1 *
global-backend

+ Add target

You can also add/remove targets after the policy is created

- Apply policy to targets (optional)

Targets are Google Cloud resources that you want to control access to. You can use external load balancer backend services as targets.

Targets

Type 1

Backend service (external application load b... ▼

Backend service 1 *
global-backend ▼ ?

+ Add target

You can also add/remove targets after the policy is created



You can also add/remove targets after the policy is created

Next step

- Advanced configurations (optional)

Adaptive protection configuration

Cloud Armor Adaptive Protection helps protect backend services from Layer 7 DDoS attacks by learning normal traffic patterns, detecting and alerting on potential attacks, and providing Cloud Armor WAF rules to mitigate them.

[Learn more](#)

☐ Enable Adaptive Protection

Content parsing configuration

Configure parsing of request body content for preconfigured WAF evaluations. [Learn more](#)

User IP request headers configuration

Add or edit request IP header name(s) to be used for User IP in evaluating Cloud Armor rules. [Learn more](#)

Request body inspection size configuration

Specify the maximum size (in KB) of the request body to inspect.

[Learn more](#)

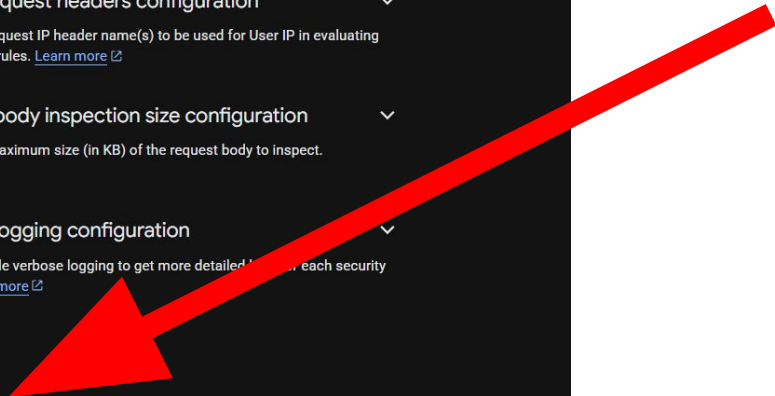
Verbose logging configuration

You can enable verbose logging to get more detailed logs for each security policy. [Learn more](#)

Done

Create policy

Cancel



Protocol forwarding deployments, or instances with public IP addresses. [Learn more](#)

Cloud Armor policies

[+ Create policy](#)

[Refresh](#)

[Filter](#) Enter property name or value

☐	Name ↑	Type	Scope	Rules	Targets ?	Description
☐	cloud-armor-week9	Backend security policy	global	1	1	cloud-
☐	default-security-policy-for-backend-service-chica-backend	Backend security policy	global	2	0	Defau
☐	default-security-policy-for-backend-service-global-backend	Backend security policy	global	2	0	Defau
☐	default-security-policy-for-backend-service-seco-friday-backend	Backend security policy	regional (us-central1)	2	0	Defau

